# Dohyun (Eric) Kim

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## EDUCATION

**University of Auckland** — **Auckland, New Zealand**
*Bachelor of Engineering (Honours) in Computer Systems Engineering* — *Expected Graduation Nov 2027*

## EXPERIENCES

**Research Assistant Trainee** — **Auckland, New Zealand**
*University of Auckland* — *Jul 2026 - Nov 2026*

- Trainee researcher analysing the **robot soccer system** at UoA's CARES Lab, covering camera calibration, **vision processing and decision strategy design**, to support its planned migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms and implemented LiDAR integration and control using **ROS and Python**, building hands-on experience in autonomous systems.

**Robotics Instructor** — **Auckland, New Zealand**
*Creative Imaginary Lab (ciLab)* — *Apr 2026 - Nov 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

## SKILLS

**Languages & Tools**: Python, C/C++, Java, JavaScript, TypeScript, SQL, R, HTML/CSS, Git, Linux, Android Studio, Figma
**Frameworks**: React, Tailwind CSS, Node.js, Express, JUnit, ROS, REST APIs, Spring Boot
**Cloud & DevOps**: AWS(Lambda, DynamoDB, S3, Bedrock, Kinesis), Docker, Terraform, GitHub Actions

## PROJECTS

**Sentiment PULSE** — **Live Demo**
*Real-time Sentiment Dashboard - Individual Project* — *Jun 2026*

- Built a serverless NLP pipeline with **AWS (Kinesis, Lambda, Comprehend, DynamoDB)** using AWS SAM, with idempotent writes, partial batch response retries, and a dead letter queue **to classify streaming text sentiment in real time** without data loss.
- Designed a **bounded DynamoDB GSI query pattern** instead of full table scans to keep read latency low, and batched Comprehend calls per Lambda invocation to cut API overhead.
- Developed a live React dashboard on AWS Amplify with **CI via GitHub Actions (21 frontend, 8 backend tests)** and a client side simulation fallback, applying cloud and AI/ML fundamentals **from certification study to a hands on project.**

**RoastWorks Analytics** — **Python**
*Business Analytics Dashboard - Team Project* — *May 2026*

- Built a PyQt6 desktop analytics app with **pandas and Matplotlib**, automating a manual Excel workflow across **3 business units** and **52,000+ records**, cutting report time from a full day to **under 30 seconds**.
- Implemented three forecasting models across **36 months of data**, combining **scikit-learn** regression with a custom **NumPy** build of Holt's Exponential Smoothing, validated with MAE and RMSE.

## AWARD

**Top 8 Finalist in AWS×BNZ AI Hackathon 2026** — **Live Demo**
*"Spottern!" - Statement Level Fraud Detection Platform* — *Jul 2026*

- Built a serverless fraud pipeline with **AWS (S3, Lambda, Textract, Bedrock, DynamoDB)**, using Claude Opus 4.8 on Bedrock **to extract and flag anomalies from bank statements** with cited transaction evidence.
- **Proposed the approach to BNZ** by drawing on prior individual project experience with serverless AWS pipelines, placing **top 8 out of 20+** teams.

## CERTIFICATIONS

**HashiCorp Certified: Terraform Associate** · HashiCorp — *Sep 2026*
**AWS Certified Solutions Architect - Associate** · Amazon Web Services — *Aug 2026*
**AWS Certified AI Practitioner** · Amazon Web Services — *May 2026*
**AWS Certified Cloud Practitioner** · Amazon Web Services — *Apr 2026*

## LEADERSHIP

**Academic Team Executive** — **Auckland, New Zealand**
*KEB (Korean Engineering Body) - Founding Member* — *Jul 2024 - Oct 2026*

- **Co-founded** an engineering community of **170+** members; contributed to a **17-member** executive team.
- Delivered **tutorial sessions** for junior engineering students and supported academic event planning.